

* Aim :- Write a java program that demonstrates array operations such as initialization, accessing elements, and performing basic array manipulation.

* Theory :- In java, an array is a data structure that is used to store multiple values of the same data type in a single variable. Arrays allow efficient storage, access, and manipulation of data elements using an index. The index of an array always starts from 0 and goes up to length - 1.

Basic array operations demonstrated :-

① Array initialization :- Array initialization is the process of assigning values to an array at the time of its declaration. In Java, this can be done directly using curly braces {}.

Ex :- `int[] numbers = { 10, 20, 30, 40, 50 };`

② Accessing Array elements :-

Array elements are accessed using their index number. For example, the first element is accessed using index 0.

Ex :- `System.out.println("First element: " + numbers[0]);`

`System.out.println("Third element: " + numbers[2]);`

③ Traversing an Array :- Traversing an array means accessing each element of the array one by one. This is usually done using loops such as the for loop.

Ex:-

```
for (int i=0; i<numbers.length; i++) {  
    System.out.println(numbers[i]);  
}
```

numbers.length gives the total number of elements in the array.

④ Array manipulation :- Array manipulation includes performing operations like :-

- (i) Finding the sum of elements
- (ii) Finding the largest element
- (iii) Updating values these operations help in understanding how data can be processed using arrays.

* Conclusion :- In this practical, we successfully studied and implemented basics of array operations in java. We learned how to initialize an array, access its elements using index values, traverse the array using loops and performing basic manipulations such as calculating the sum and finding largest element.